Sprint 2 – Requirements and Design

Student Result Management System for Educational Institutions

Team Members

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1. Introduction

This report provides the deliverables of Sprint 2 of the Student Result Management System (SRMS) web application. This sprint has the purpose of specifying a complete list of user requirements in the form of user stories, and designing the first components of the application, such as personas and scenarios, a use case model, wireframes, and user flows. The Student Result Management System is a comprehensive academic performance tracking and result publishing platform developed for educational institutions, built using Python (Django/Flask), PostgreSQL, HTML/CSS, and version managed with Git.

The system enables educational institutions to manage student academic records, publish results, generate transcripts, and provide role-based access to administrators, faculty, and students. The platform replaces manual paper-based result processing with a secure, efficient, and scalable digital solution.

2. User Stories

These are the functional requirements of the first version of the Student Result Management System, which are specified as the following user stories. They are provided from the perspective of the institution administrator, faculty members, and students as the primary users of the system. These user stories are also tracked on the team task board with the corresponding labels and priorities.

1. As an administrator, I wish to gain secure access to the system to ensure only authorised personnel can manage student records and results.

2. As a faculty member, I would like to be able to view a dashboard displaying the list of subjects I am assigned to, along with their pending result entry deadlines.

3. As a faculty member, I wish to enter marks for students in my assigned subjects so that their academic performance is recorded accurately.

4. As a faculty member, I desire to edit previously submitted marks before the result is published so that I can correct data entry errors in a timely manner.

5. As a faculty member, I would like to import student marks from a CSV file so that I can save time on bulk data entry for large batches.

6. As an administrator, I wish to manage the academic calendar by creating and updating semesters and examination sessions so that results are organised by the correct time periods.

7. As an administrator, I would like to register new students and assign them to a programme and semester so that they are associated with the correct academic structure.

8. As an administrator, I would like to enrol students in specific subjects for a semester so that only the enrolled students appear in the marks entry sheets.

9. As an administrator, I wish to configure pass marks, grade boundaries, and grading scales per subject or programme so that the system computes grades automatically.

10. As an administrator, I would like to publish results for a specific semester so that students can view their grades upon official release.

11. As an administrator, I wish to withhold the result of a specific student (e.g. due to fee arrears) so that only eligible students can access their results.

12. As a student, I would like to log in and view my published results for any completed semester so that I can track my academic progress.

13. As a student, I wish to download my official result transcript in PDF format so that I can submit it for applications or record-keeping.

14. As a student, I would like to see a graphical summary of my GPA trends over multiple semesters so that I can visually track my performance over time.

15. As an administrator, I would like to search for any student by name, registration number, or programme so that I can quickly find and manage their records.

16. As an administrator, I wish to generate class-level result sheets for a subject showing all students, their marks, grades, and pass/fail status for official reporting.

17. As an administrator, I would like to track and view the academic history of individual students across all semesters in a single profile view.

18. As an administrator, I would like to receive automated alerts when a faculty member has not submitted marks by the deadline so that I can take follow-up action.

19. As an administrator, I wish to generate statistical reports showing the pass rate, fail rate, and grade distribution for a subject or department so that I can support academic quality reviews.

20. As an administrator, I would like to export result data in Excel or CSV format so that it can be shared with external bodies or further analysed offline.

21. As a faculty member, I would like to view the class performance analysis for my subject after marks are entered so that I can identify students who may need academic support.

22. As an administrator, I wish to manage user accounts for faculty and staff, including creating, deactivating, and resetting passwords so that system access is always controlled.

23. As a student, I would like to receive a notification when my results are published so that I am promptly informed without having to check the portal repeatedly.

24. As an administrator, I wish the system to validate all marks entries to ensure they fall within the maximum marks allowed for each subject to prevent erroneous data from being saved.

3. User Stories and Task Board Tracking

Each of the above user stories is placed on the team task board and labelled with the respective epic or category (Authentication, Results Entry, Student Management, Programme & Semester Management, Reporting, Notifications). The user stories are assigned a priority and allocated to a sprint. Stories are tracked across columns — Backlog, To Do, In Progress, In Review, and Done — to monitor progress throughout the development lifecycle.

Epic / Category	User Story IDs	Priority
Authentication & Access Control	1, 22	High
Results Entry & Grading	2, 3, 4, 5, 9, 24	High
Student Management	7, 8, 11, 15, 17	High
Programme & Semester Management	6, 10	Medium
Student Self-Service Portal	12, 13, 14, 23	Medium
Reporting & Analytics	16, 19, 20, 21	Medium
Notifications & Alerts	18, 23	Low

4. Personas and Scenarios

Persona 1 – Institution Administrator

Name: Lucas Mathew

Age: 34

Role: Academic Registrar / System Administrator

The administrator oversees the entire academic records management process for the institution. They are proficient with web-based administrative systems and require a centralised platform to manage students, faculty, academic sessions, and result publication with minimal manual effort.

Scenario:

At the start of a new examination session, the administrator logs into Student Result Management System and creates a new semester under the current academic year. They enrol the relevant student cohorts into their respective subjects and assign faculty members as result owners for each subject. As the submission deadline approaches, the administrator checks the dashboard for pending submissions and sends automated reminders to faculty who have not yet uploaded marks. Once all marks are received, the administrator reviews the grade distribution report, resolves any flagged anomalies, and officially publishes the results. Students are automatically notified. The administrator then generates and archives the official result sheets for institutional records.

Persona 2 – Faculty Member

Name: Mohammad Ahmad

Age: 37

Role: Lecturer / Subject Coordinator

The faculty member is responsible for delivering a subject and submitting student assessment results by the stipulated deadline. They are comfortable with digital tools and prefer a quick and reliable way to enter or upload marks and review class performance after submission.

Scenario:

After the end-of-semester examinations, the faculty member logs into SRMS and navigates to the Result Entry page for their assigned subject. They choose to upload a pre-prepared CSV file with student registration numbers and marks. The system validates the entries and flags one student whose mark exceeds the maximum allowed. The faculty member corrects the error and resubmits. The system automatically computes the grades and pass/fail status based on the pre-configured grading scale. The faculty member then reviews the class performance summary showing the average score and grade distribution before submitting the results to the administrator for approval.

Persona 3 – Student

Name: Marshal Darius

Age: 20

Role: Undergraduate Student

The student uses the system to view published results, monitor their GPA trends, and download transcripts for external use. They value transparency, timely result availability, and easy access to their academic history from any device.

Scenario:

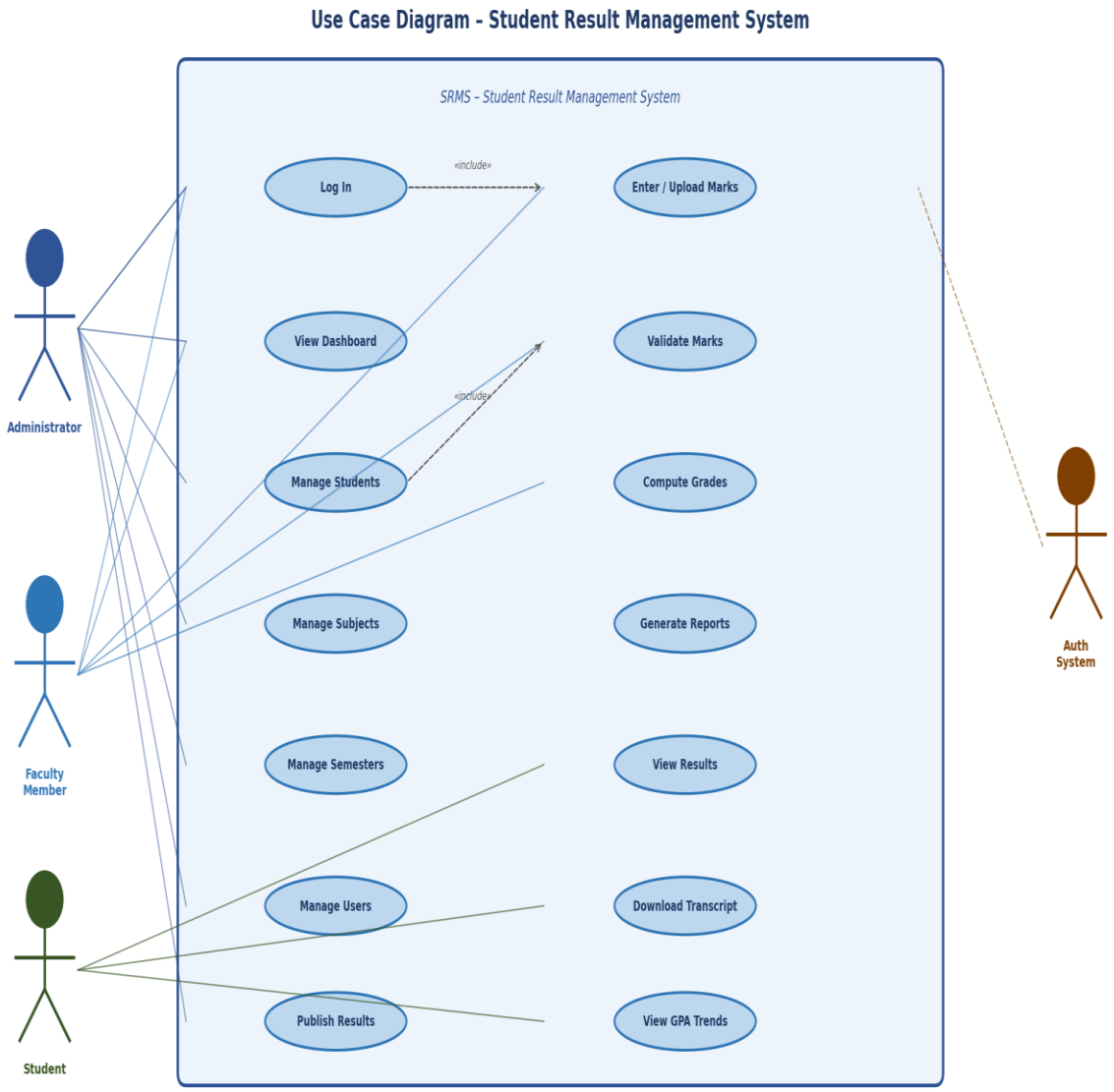
On the day of result publication, the student receives an email notification and logs into the SRMS student portal. They navigate to the Results section and view their grades for the current semester displayed as a structured result card showing subject names, marks obtained, maximum marks, grades, and the semester GPA. Noticing their GPA has improved compared to the previous semester, the student views the GPA trend chart over all semesters. They then download a PDF transcript to attach to a scholarship application.

5. Use Case Model

The use case model of the Student Result Management System is described through a use case diagram and descriptions of the key use cases.

Actors:

- Institution Administrator
- Faculty Member
- Student
- Authentication System



The key use cases are: Log In, Dashboard, Student Management, Programme & Semester Management, Result Entry, Result Publication, Transcript Generation, and Reporting. The Authentication System actor is associated with the Log In use case, which all human actors trigger as the entry point to the system.

The Faculty Member actor is linked to View Dashboard, Enter / Upload Marks, Edit Pending Marks, and View Class Performance Analysis use cases. The Administrator actor is associated with all use cases including Manage Students, Manage Users, Manage Subjects, Manage Semesters, Publish Results, Withhold Results, and Generate Reports. The Student actor is linked to View Results, Download Transcript, and View GPA Trends.

The Manage Students use case generalises the operations to register, search, update, and enrol students. Similarly, the Result Entry use case encompasses entering marks

manually, importing from CSV, and validating entries. The Generate Reports use case covers class result sheets, grade distribution statistics, and exportable data outputs.

6. Wireframes

The wireframes define the layout and content of the major screens of the application. They follow a consistent design language with a persistent top navigation bar and sidebar for role-specific actions.

Login Page

A centred login form with the application logo and name at the top. Two input fields are provided for username/email and password, followed by a Login button. A Forgot Password link appears below the form. An error message area is displayed beneath the form on invalid login attempts.

Administrator Dashboard

The dashboard presents a top horizontal navigation bar with links to Dashboard, Students, Subjects, Semesters, Results, Reports, and User Management, along with a logout option. The main content area shows summary cards displaying the total number of students, active subjects, upcoming result deadlines, and pending faculty submissions. A table below lists the current examination sessions with their result submission status.

Result Entry Page (Faculty)

The result entry page shows a form header with the subject name, semester, and maximum marks. Below the header is a table populated with all enrolled students for the subject, containing columns for registration number, student name, marks obtained, computed grade (read-only), and a status indicator. A button at the top allows importing data from CSV. Save Draft and Submit for Approval buttons appear at the bottom of the table.

Student Result View Page

The student portal result page displays a result card for the selected semester showing the institution name, student name, registration number, programme, and semester. A table lists each subject with columns for subject code, subject name, marks, maximum marks, grade, and credit hours. The semester GPA and total credits are shown at the bottom of the card. A Download PDF button is prominently placed at the top right.

Reports Page (Administrator)

The reports page provides a date range filter and a dropdown to select the subject or department. Options to generate Class Result Sheet, Grade Distribution Chart, Staff Submission Summary, and Export to Excel are displayed as action buttons. Generated reports appear inline below the controls in a tabular or chart format.

Student Profile Page

The student profile page shows the student's photograph, name, registration number, programme, and enrolment year on a card on the left. The right panel contains a tabbed

view with tabs for Academic History, Enrolled Subjects, and Personal Details. The Academic History tab shows a list of all semesters with their GPA values and a line chart showing GPA trends over time.

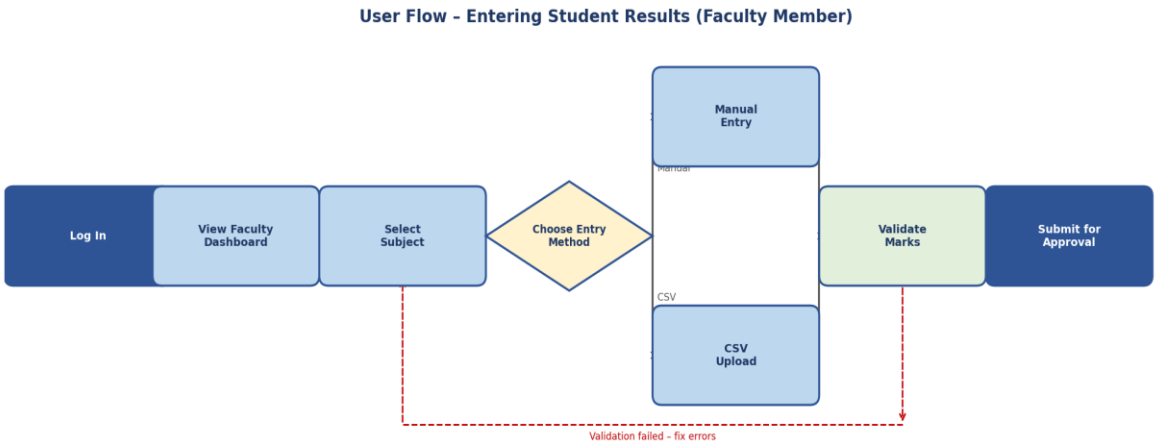
7. User Flow Diagrams

The user flows describe step-by-step how users navigate the system to complete key tasks.

Flow 1: Entering Student Results (Faculty Member)

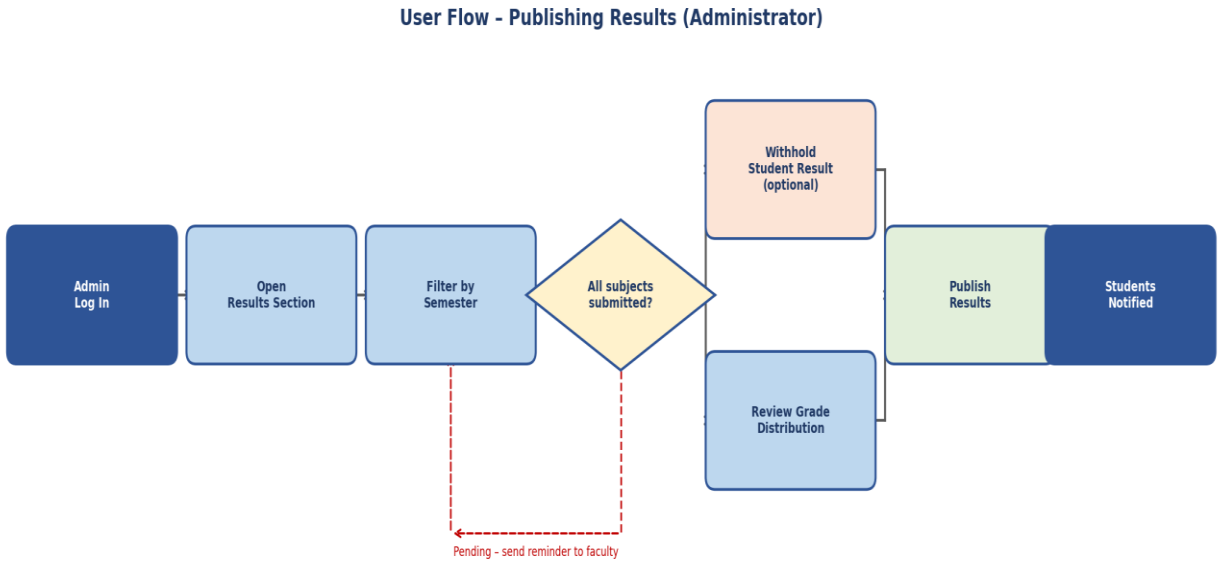
The faculty member logs into the system and is directed to their dashboard. They select a subject from the list of assigned subjects. On the subject result entry page, they choose to either enter marks manually row by row or upload a CSV file. If uploading, the system parses the file and displays a preview with validation errors highlighted. The faculty member corrects any flagged entries and proceeds. Once all marks are entered, the system automatically calculates the grades based on the configured grading scale. The faculty member reviews the class performance summary and clicks Submit for

Approval. The system saves the results with a pending status and notifies the administrator.



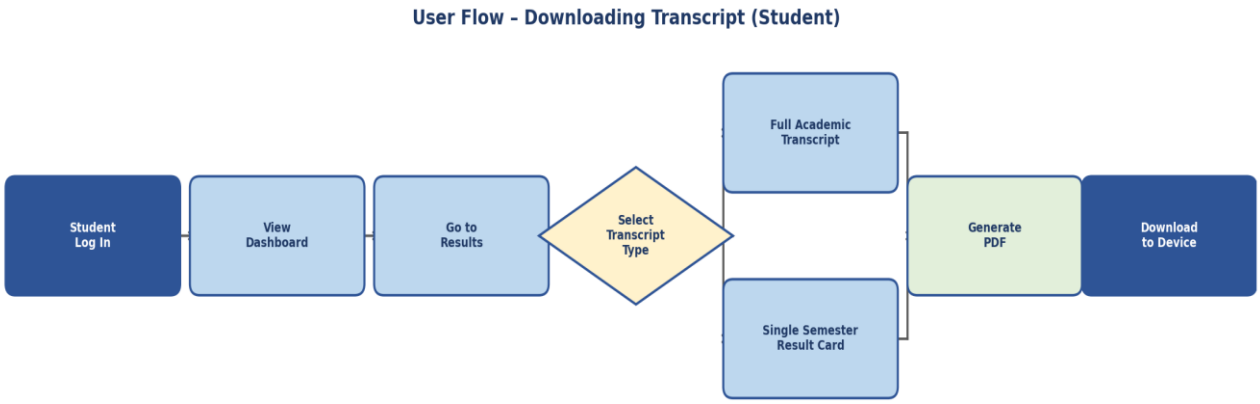
Flow 2: Publishing Results (Administrator)

The administrator logs in and navigates to the Results section. They filter by semester to see all subjects and their submission statuses. For subjects where marks have been submitted for approval, the administrator reviews the entries and the grade distribution summary. If the data is satisfactory, the administrator clicks Publish Results for the selected semester. The system changes the result status to Published, making them visible to students. All affected students receive an automated notification. If the administrator needs to withhold a student's result, they locate the student record and toggle the result visibility flag to hidden before publishing the semester results.



Flow 3: Downloading a Transcript (Student)

The student logs into the portal and is directed to their dashboard. They navigate to the Results section and select the Transcript option. The system presents all completed semesters with an option to generate a full academic transcript or a single-semester result card. The student selects Full Transcript and clicks Download PDF. The system generates a formatted PDF containing all semester results, cumulative GPA, and the institution's official header and footer, which is then downloaded to the student's device.



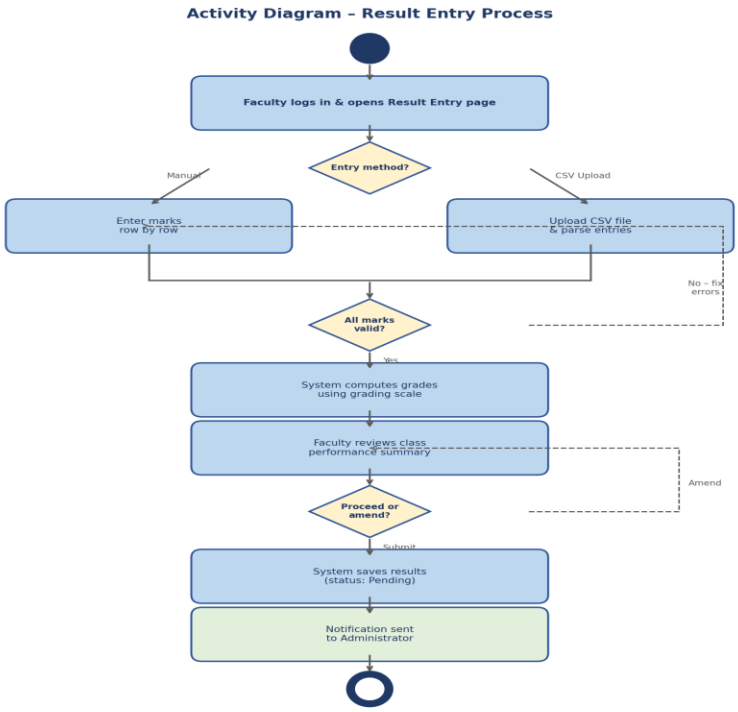
Flow 4: Generating a Report (Administrator)

The administrator logs in and selects Reports from the navigation bar. They specify a semester and optionally a subject or department. The system queries the database and renders a grade distribution report displaying pass rates, fail rates, average scores, and a breakdown by grade band. The administrator can toggle between a bar chart view and a tabular view. They then click Export to Excel to download the data for submission to the academic board.

8. Optional Activity and Sequence Diagrams

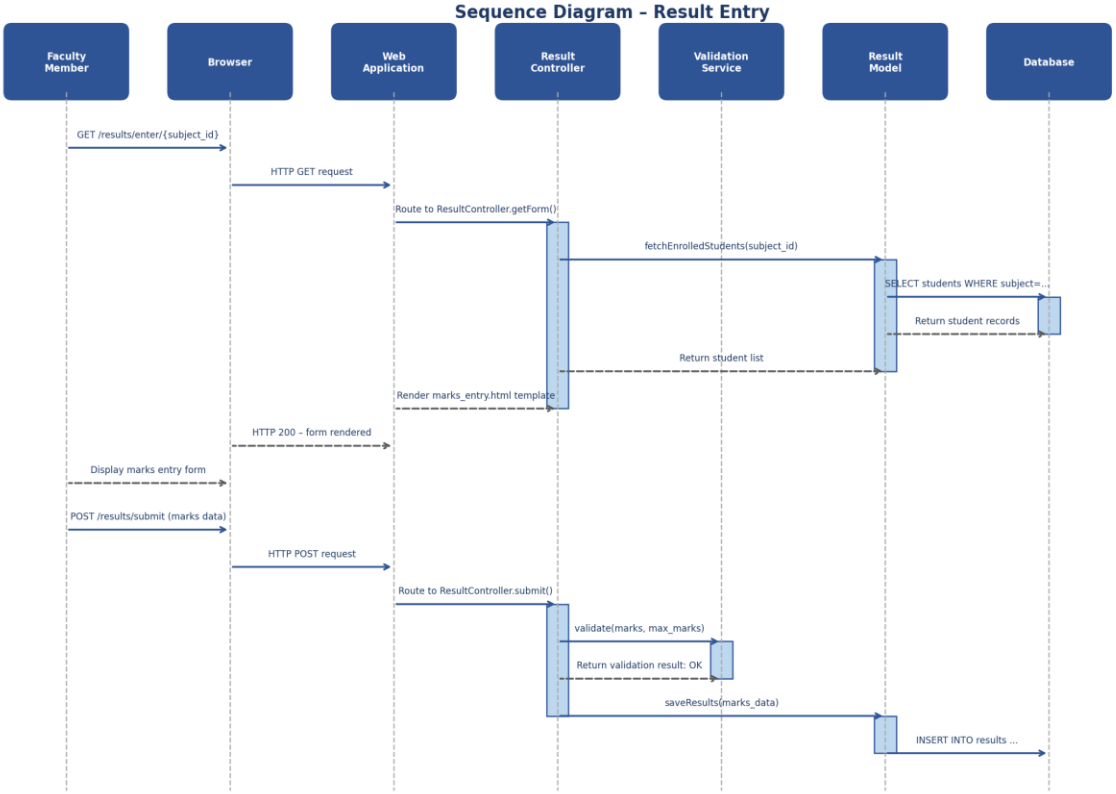
Activity Diagram: Result Entry Process

The result entry activity begins with the faculty member initiating the result entry for a subject on the Result Entry page. The next step is choosing the entry method: manual entry or CSV upload. For manual entry, the faculty member fills in the marks row by row. For CSV upload, the system parses the file and validates each row. A decision node evaluates whether all marks are within the allowed maximum. If validation fails for any record, the flow returns to the entry step with error messages highlighted on the offending rows. Once all marks pass validation, the system computes grades automatically using the configured grading scale. The faculty member reviews the class performance summary, and a decision node checks if they proceed or require further amendments. If the faculty member submits, the system saves the results with a Pending status and triggers a notification to the administrator.



Sequence Diagram: Result Entry

The components involved in the result entry sequence diagram are the Faculty Member, Browser, Web Application, Result Controller, Result Model, Marks Validation Service, and Database. The faculty member initiates a GET request through the browser to load the result entry form for a given subject. The browser sends an HTTP request to the web application, which is routed to the Result Controller. The controller queries the Result Model for the list of enrolled students, which in turn queries the Database and returns the records. The controller renders the marks entry template and the browser displays the form to the faculty member.

Upon form submission, the browser sends a POST request containing marks data. The Result Controller passes the data to the Marks Validation Service, which checks each entry against the maximum marks configured for the subject. If validation passes, the controller calls the Result Model to persist the marks records to the Database. The database confirms the write operation. The controller responds to the browser with a redirect to the class performance summary page, and the system records the submission status as Pending.

Sequence Diagram – Result Entry